

```
akshi@akshita:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
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This is free software; see the source for copying conditions. There is
NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

```
akshi@akshita:~$ make --version
GNU Make 4.4.1
Built for x86_64-pc-linux-gnu
Copyright (C) 1988-2023 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
```

```
akshi@akshita:~$ git --version
git version 2.53.0
```

```
akshi@akshita:~$ strace --version
strace -- version 6.19
Copyright (c) 1991-2026 The strace developers <https://strace.io>.
This is free software; see the source for copying conditions. There is
NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

```
Optional features enabled: stack-trace=libunwind stack-demangle m32-mpers
s mx32-mpers
```

```
akshi@akshita:~$ mkdir ossp-shell
```

```
akshi@akshita:~$ cd ossp-shell
```

```
akshi@akshita:~/ossp-shell$ git init
```

```
hint: Using 'master' as the name for the initial branch. This default br
anch name
```

```
hint: will change to "main" in Git 3.0. To configure the initial branch
name
```

```
hint: to use in all of your new repositories, which will suppress this w
arning,
```

```
hint: call:
```

```
hint: Using 'master' as the name for the initial branch. This default br
anch name
hint: will change to "main" in Git 3.0. To configure the initial branch
name
hint: to use in all of your new repositories, which will suppress this w
arning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this com
mand:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
```

```
Initialized empty Git repository in /home/akshi/oss-shell/.git/
```

```
akshi@akshita:~/oss-shell$ pwd
```

```
/home/akshi/oss-shell
```

```
akshi@akshita:~/oss-shell$ mkdir src include bin build
```

```
akshi@akshita:~/oss-shell$ touch src/main.c src/shell.c include/shell.h
Makefile
```

```
akshi@akshita:~/oss-shell$ tree
```

```
.
├── Makefile
├── bin
├── build
├── include
│   └── shell.h
└── src
    ├── main.c
    └── shell.c
```

```
5 directories, 4 files
```

```
akshi@akshita:~/oss-shell$ nano src/main.c
```

```
akshi@akshita:~/oss-shell$ gcc src/main.c -o bin/test
```

```
akshi@akshita:~/ossp-shell$ ./bin/test
OSSP Shell Project Started!
akshi@akshita:~/ossp-shell$ ls -l bin
total 16
-rwxr-xr-x 1 akshi akshi 15952 Aug  8 09:53 test
akshi@akshita:~/ossp-shell$ nano Makefile
akshi@akshita:~/ossp-shell$ make
gcc -Wall -Wextra -Iinclude -c src/main.c -o src/main.o
gcc -Wall -Wextra -Iinclude -c src/shell.c -o src/shell.o
gcc -Wall -Wextra -Iinclude src/main.o src/shell.o -o bin/myshell
akshi@akshita:~/ossp-shell$ ls -l bin
total 32
-rwxr-xr-x 1 akshi akshi 15984 Aug  8 09:55 myshell
-rwxr-xr-x 1 akshi akshi 15952 Aug  8 09:53 test
akshi@akshita:~/ossp-shell$ make clean
rm -f src/main.o src/shell.o bin/myshell
akshi@akshita:~/ossp-shell$ ls bin
test
akshi@akshita:~/ossp-shell$ make
gcc -Wall -Wextra -Iinclude -c src/main.c -o src/main.o
gcc -Wall -Wextra -Iinclude -c src/shell.c -o src/shell.o
gcc -Wall -Wextra -Iinclude src/main.o src/shell.o -o bin/myshell
akshi@akshita:~/ossp-shell$ ls bin
myshell test
akshi@akshita:~/ossp-shell$ ./bin/myshell
OSSP Shell Project Started!
akshi@akshita:~/ossp-shell$ nano src/fork_demo.c
akshi@akshita:~/ossp-shell$ gcc -Wall -Wextra src/fork_demo.c -o bin/fork_demo
akshi@akshita:~/ossp-shell$ ./bin/fork_demo
Before fork()
Parent process
Parent PID = 11662
Child process
Child PID = 11663
Child PID = 11663
akshi@akshita:~/ossp-shell$ ^C
akshi@akshita:~/ossp-shell$ nano src/exec_demo.c
akshi@akshita:~/ossp-shell$ gcc -Wall -Wextra src/exec_demo.c -o bin/exe
```

```
c_demo
akshi@akshita:~/ossp-shell$ ./bin/exec_demo
Before exec()
total 20
-rw-r--r-- 1 akshi akshi 294 Aug 8 09:55 Makefile
drwxr-xr-x 2 akshi akshi 4096 Aug 8 09:58 bin
drwxr-xr-x 2 akshi akshi 4096 Aug 8 09:52 build
drwxr-xr-x 2 akshi akshi 4096 Aug 8 09:52 include
drwxr-xr-x 2 akshi akshi 4096 Aug 8 09:58 src
akshi@akshita:~/ossp-shell$ execlp("ls", "ls", "-l", NULL);
-bash: syntax error near unexpected token `"ls",'
akshi@akshita:~/ossp-shell$ nano src/exec_demo.c
akshi@akshita:~/ossp-shell$ nano src/fork_exec.c
akshi@akshita:~/ossp-shell$ gcc -Wall -Wextra src/fork_exec.c -o bin/fork_exec
akshi@akshita:~/ossp-shell$ ^C
akshi@akshita:~/ossp-shell$ ./bin/fork_exec
Parent: waiting for child
Child: executing ls
total 20
-rw-r--r-- 1 akshi akshi 294 Aug 8 09:55 Makefile
drwxr-xr-x 2 akshi akshi 4096 Aug 8 10:00 bin
drwxr-xr-x 2 akshi akshi 4096 Aug 8 09:52 build
drwxr-xr-x 2 akshi akshi 4096 Aug 8 09:52 include
drwxr-xr-x 2 akshi akshi 4096 Aug 8 10:00 src
Parent: child finished
akshi@akshita:~/ossp-shell$ nano src/shell.c
akshi@akshita:~/ossp-shell$ nano include/shell.h
akshi@akshita:~/ossp-shell$ nano src/main.c
akshi@akshita:~/ossp-shell$ make clean
rm -f src/main.o src/shell.o bin/myshell
akshi@akshita:~/ossp-shell$ make
gcc -Wall -Wextra -Iinclude -c src/main.c -o src/main.o
gcc -Wall -Wextra -Iinclude -c src/shell.c -o src/shell.o
gcc -Wall -Wextra -Iinclude src/main.o src/shell.o -o bin/myshell
akshi@akshita:~/ossp-shell$ ./bin/myshell
myshell$ pwd
/home/akshi/ossp-shell
myshell$ whoami
```

```
myshell$ whoami
akshi
myshell$ date
Sat Aug 8 10:03:52 UTC 2026
myshell$ ls
Makefile bin build include src
myshell$ exit
akshi@akshita:~/ossp-shell$ nano src/shell.c
akshi@akshita:~/ossp-shell$ make clean
rm -f src/main.o src/shell.o bin/myshell
akshi@akshita:~/ossp-shell$ make
gcc -Wall -Wextra -Iinclude -c src/main.c -o src/main.o
gcc -Wall -Wextra -Iinclude -c src/shell.c -o src/shell.o
gcc -Wall -Wextra -Iinclude src/main.o src/shell.o -o bin/myshell
akshi@akshita:~/ossp-shell$ ./bin/myshell
myshell$ myshell$ ls -l
execvp: No such file or directory
myshell$ ls -l
total 20
-rw-r--r-- 1 akshi akshi 294 Aug 8 09:55 Makefile
drwxr-xr-x 2 akshi akshi 4096 Aug 8 10:05 bin
drwxr-xr-x 2 akshi akshi 4096 Aug 8 09:52 build
drwxr-xr-x 2 akshi akshi 4096 Aug 8 10:01 include
drwxr-xr-x 2 akshi akshi 4096 Aug 8 10:05 src
myshell$ ls -a
. .. .git Makefile bin build include src
myshell$ echo helo world
helo world
myshell$ pwd
/home/akshi/ossp-shell
myshell$ exit
akshi@akshita:~/ossp-shell$ nano src/shell.c
akshi@akshita:~/ossp-shell$ make clean
rm -f src/main.o src/shell.o bin/myshell
akshi@akshita:~/ossp-shell$ make
gcc -Wall -Wextra -Iinclude -c src/main.c -o src/main.o
gcc -Wall -Wextra -Iinclude -c src/shell.c -o src/shell.o
gcc -Wall -Wextra -Iinclude src/main.o src/shell.o -o bin/myshell
akshi@akshita:~/ossp-shell$ ./bin/myshell
```

```

gcc -Wall -Wextra -Iinclude src/main.o src/shell.o -o bin/myshell
akshi@akshita:~/ossp-shell$ ./bin/myshell
myshell$ echo Hello World
Hello World
myshell$ echox
echox
execvp: No such file or directory
myshell$ ./bin/myshell
./bin/myshell
myshell$ echox
execvp: No such file or directory
myshell$ hello
hello
execvp: No such file or directory
myshell$ echo hello
echo hello
hello
myshell$ echo Operating Systems And Systems Programming
echo Operating Systems And Systems Programming
Operating Systems And Systems Programming
myshell$ exit
exit
myshell$ exit
exit
akshi@akshita:~/ossp-shell$ ./bin/myshell
myshell$ pwd
/home/akshi/ossp-shell
myshell$ exit
exit
akshi@akshita:~/ossp-shell$ pstree -p
systemd(1)─agetty(239)
            └─chronyd-starter(139)─chronyd(295)─chronyd(313)
            └─cron(140)
            └─dbus-daemon(141)
            └─init-systemd(Ub(2)─SessionLeader(892)─Relay(897)(893)─+
++

```

```

OST|ONLCK, c_cflag=B50400|CS0|CREAD, c_cflag=1310|ICANON|ECHO|ECHOL|EC
OK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
kit_group(0) = ?
++ exited with 0 +++
akshi@akshita:~/ossp-shell$ pstree -p
systemd(1)─agetty(239)
          └─chronyd-starter(139)─chronyd(295)─chronyd(313)
          └─cron(140)
          └─dbus-daemon(141)
          └─init-systemd(Ub(2)─SessionLeader(892)─Relay(897)(893)─+
+
          │
          │   └─init(6)─{init}(7)
          │   └─login(376)─bash(452)
          │   └─{init-systemd(Ub}(8)
          └─networkd-dispat(149)
          └─polkitd(1041)─{polkitd}(1043)
                        └─{polkitd}(1044)
                        └─{polkitd}(1045)
          └─rsyslogd(4807)─{rsyslogd}(4821)
                        └─{rsyslogd}(4822)
                        └─{rsyslogd}(4823)
          └─systemd(420)─(sd-pam)(422)
                        └─dbus-daemon(9682)
          └─systemd-journal(4948)
          └─systemd-logind(158)
          └─systemd-resolve(5859)
          └─systemd-udevd(5482)
          └─unattended-upgr(303)─{unattended-upgr}(331)

```

```
akshi@akshita:~/oss-sp-shell$ git config --global user.name "Akshita"
akshi@akshita:~/oss-sp-shell$ git config --global user.email "aakshita608@gmail.com"
akshi@akshita:~/oss-sp-shell$ git config --global user.name
Akshita
akshi@akshita:~/oss-sp-shell$ git config --global user.email
aakshita608@gmail.com
akshi@akshita:~/oss-sp-shell$ git commit -m "Complete OSSP shell assignment"
[master (root-commit) 8a48436] Complete OSSP shell assignment
14 files changed, 287 insertions(+)
create mode 100644 Makefile
create mode 100755 bin/exec_demo
create mode 100755 bin/fork_demo
create mode 100755 bin/fork_exec
create mode 100755 bin/myshell
create mode 100755 bin/test
create mode 100644 include/shell.h
create mode 100644 src/exec_demo.c
create mode 100644 src/fork_demo.c
create mode 100644 src/fork_exec.c
create mode 100644 src/main.c
create mode 100644 src/main.o
create mode 100644 src/shell.c
create mode 100644 src/shell.o
akshi@akshita:~/oss-sp-shell$ git status
On branch master
nothing to commit, working tree clean
akshi@akshita:~/oss-sp-shell$ git log --oneline -1
8a48436 (HEAD -> master) Complete OSSP shell assignment
akshi@akshita:~/oss-sp-shell$ make clean
make
rm -f src/main.o src/shell.o bin/myshell
gcc -Wall -Wextra -Iinclude -c src/main.c -o src/main.o
gcc -Wall -Wextra -Iinclude -c src/shell.c -o src/shell.o
gcc -Wall -Wextra -Iinclude src/main.o src/shell.o -o bin/myshell
akshi@akshita:~/oss-sp-shell$ ./bin/myshell
myshell$ pwd
```



```
akshi@akshita:~/ossp-shell$ make clean
make
rm -f src/main.o src/shell.o bin/myshell
gcc -Wall -Wextra -Iinclude -c src/main.c -o src/main.o
gcc -Wall -Wextra -Iinclude -c src/shell.c -o src/shell.o
gcc -Wall -Wextra -Iinclude src/main.o src/shell.o -o bin/myshell
akshi@akshita:~/ossp-shell$ ./bin/myshell
myshell$ pwd
/home/akshi/ossp-shell
myshell$ ls
ls
Makefile  bin  build  include  src
myshell$ ls -l
ls -l
total 20
-rw-r--r-- 1 akshi akshi 294 Aug  8 09:55 Makefile
drwxr-xr-x 2 akshi akshi 4096 Aug  9 13:30 bin
drwxr-xr-x 2 akshi akshi 4096 Aug  8 09:52 build
drwxr-xr-x 2 akshi akshi 4096 Aug  8 10:01 include
drwxr-xr-x 2 akshi akshi 4096 Aug  9 13:30 src
myshell$ echo hello world
echo hello world
hello world
myshell$ date
date
Sun Aug  9 13:30:34 UTC 2026
myshell$ whoami
whoami
akshi
myshell$ echo
echo

myshell$ hello
hello
execvp: No such file or directory
myshell$ exit
exit
akshi@akshita:~/ossp-shell$
```